

Tory Brook Catchment Delineation & Storm Response Analysis

A civil engineering GIS portfolio project- Tory Brook sub-catchment, River Plym, Devon

Data Sources

- LIDAR Composite DTM (1m/2m resolution): Environment Agency (environment.data.gov.uk/survey)
- OS Terrain 50: Ordnance Survey, used as a wider-context reference layer
- OpenStreetMap: base map layer for the final output
- EA Flood-Monitoring API(<https://environment.data.gov.uk/flood-monitoring/id/stations>): real-time and historic 15-minute rainfall and river level readings as well as EA daily archive CSV dumps for historic dates
- Flood alert records: Plymouth City Council / Environment Agency public notices

Tools

- QGIS (3.44) as the primary GIS environment
- GRASS GIS tools (r.fill.dir, r.watershed and r.water.outlet) for hydrological DEM processing and catchment delineation
- SAGA GIS tools (Fill Sinks, Channel Network) for DEM conditioning and stream network extraction
- Python/pandas for cleaning, aligning, analysing rainfall and river level time series

Key Steps

1. Scoping: selected the Tory Brook sub-catchment instead of the full 54 km² River Plym catchment for the project to be more manageable
2. DEM preparation: merged LIDAR tiles covering the study area, clipped to the catchment extent as well as hydrologically conditioned (filled sinks) in order to remove artificial depressions.
3. Flow modelling: computed flow direction and flow accumulation using GRASS's r.watershed.
4. Catchment delineation: used r.water.outlet with the surveyed gauge coordinates (NGR SX 55150 57654) to trace the true upstream contributing area, producing a single clean catchment boundary of approximately 15 km².

5. Stream network extraction: thresholded the flow accumulation raster and extracted the channel network using SAGA's Channel Network tool.
6. Validation: cross-checked the pour point location against the EA's official gauge grid reference, confirming correct placement to within a few centimetres.
7. Rainfall–level response analysis: pulled 15 minute rainfall and river level data for a documented storm event (11 to 12 November 2025, associated with a real EA flood alert for the River Plym, Tory Brook and Long Brook), cleaned and time aligned the series as well as computed lag times between rainfall and level peaks.
8. Final cartographic output: composed a final A3 map showing the catchment boundary, stream network, hillshade, pour point and base map, with full cartographic elements (title, north arrow, scale bar, legend and data source credits).

Key Findings

- Catchment area: approximately 15 km², delineated from the surveyed Newnham Park gauge location.
- Storm response (11–12 Nov 2025 event): 29.5 mm of rainfall over two days, in two pulses. Newnham Park (Tory Brook) peaked 1h 15min after peak rainfall intensity; the larger Carnwood gauge (River Plym) peaked 2h 45min after. This is consistent with the catchment scale governing response time (smaller catchments respond faster).
- Validation against a real flood alert: the analysed storm event aligned with an EA flood alert issued the evening before, confirming the delineated catchment and response timing are physically plausible.
- Methodological note: an initial storm event (4 April 2026) was found to have negligible rainfall and level responses inconsistent with genuine hydrological signal and was discarded in favour of the documented November event. Illustrating the importance of storm event selection in this kind of analysis.

Reflections

This project involved substantial troubleshooting of GIS raster/vector workflows. This includes diagnosing and correcting an initial catchment delineation error caused by an insufficiently large DEM extent (identified via inspection of GRASS flow accumulation edge-effect values) and working through tool specific quirks between SAGA and GRASS data formats. These issues and their resolutions, are documented as part of the working process behind the final deliverables.